GSoC 2026 Project Proposal

Matrix Product State Representations of Gaussian and Non-Gaussian Quantum States

Organization: Julia

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1 Introduction and Contributor Information

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2 Abstract

Simulating quantum many-body systems is a central challenge due to the exponential scaling of Hilbert space: for N particles with local dimension d, the full space dimension is d^N, rendering exact simulation intractable for $N \gtrsim 30$.

Matrix Product States (MPS) provide an efficient representation for low-entanglement states:

$$|\psi\rangle = \sum_{i_1,\ldots,i_N} \sum_{\alpha_1,\ldots,\alpha_{N-1}} A^{[1]i_1}_{1\alpha_1} A^{[2]i_2}_{\alpha_1\alpha_2} \cdots A^{[N]i_N}_{\alpha_{N-1}1} |i_1, \ldots, i_N\rangle,$$

with bond dimension $\chi = \max_k \chi_k$ giving $\mathcal{O}(Nd\chi^2)$ parameters—exponentially smaller than d^N.

While MPS methods are established for discrete-variable systems, extending them to **continuous-variable (CV) quantum systems** remains active. This project implements a unified framework for Gaussian and non-Gaussian CV states using MPS, building on Yanagimoto et al. (2021) [1] and Nüßeler et al. (2021) [2].

3 Mathematical Preliminaries

3.1 Continuous-Variable Systems

For a bosonic mode, quadrature operators $\hat{q} = (\hat{a} + \hat{a}^\dagger)/\sqrt{2}$, $\hat{p} = (\hat{a} - \hat{a}^\dagger)/(i\sqrt{2})$ satisfy $[\hat{q}, \hat{p}] = i$. Phase space is $\mathbb{R}^{2N}$ with symplectic form $\Omega = \bigoplus_{k=1}^{N} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$.

3.2 Gaussian States

A Gaussian state has characteristic function $\chi(\xi) = \exp(-\frac{1}{4}\xi^\top \Omega\sigma\Omega^\top \xi + i\mathbf{d}^\top \Omega\xi)$, with displacement $\mathbf{d} = \langle \hat{\mathbf{r}} \rangle$ and covariance $\sigma_{ij} = \frac{1}{2}\langle\{\hat{r}_i, \hat{r}_j\}\rangle - \langle \hat{r}_i \rangle\langle \hat{r}_j \rangle$.

3.3 Bogoliubov Transformations

A Bogoliubov transformation T preserves CCR: $T\Omega T^\dagger = \Omega$. The Bloch-Messiah decomposition [2] gives $T = \bar{U}\bar{S}\bar{V}^\dagger$ with passive transformations $\bar{U}, \bar{V}$ and active squeezing $\bar{S} = \begin{pmatrix} \cosh\Theta & \sinh\Theta \\ \sinh\Theta & \cosh\Theta \end{pmatrix}$.

3.4 Supermode Demultiplexing [1]

For supermodes $\hat{A}^{(r)} = \sum_m f_m^{(r)} \hat{a}_m$, there exists unitary $\hat{V}$ (beam splitters + phase shifters) such that $\hat{V}^\dagger \hat{A}^{(r)} \hat{V} = \hat{a}_r$. Angles determined iteratively:

$$\theta_m^{(r)} = \arg(I_m), \quad \phi_m^{(r)} = \sin^{-1}|I_m|, \quad I_m = \frac{f_{m-r+1}^{(r)} - \sum_{k=r}^{m-1} g_k^{(r)} c_{k,m-r+1}^{(r-1)}}{c_{m,m-r+1}^{(r-1)} \prod_{k=r}^{m-1} \cos\phi_k^{(r)}}.$$

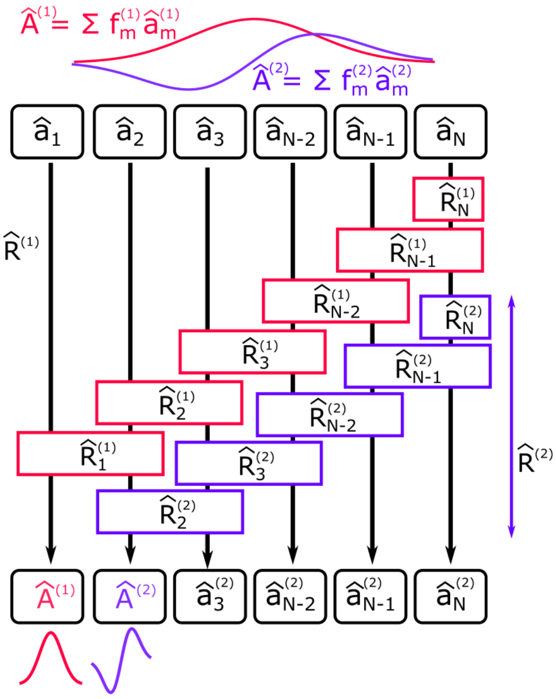

Figure 1: Illustration of the supermode demultiplexing scheme. Application of unitary transformation $\hat{R}^{(r)}$, composed of one-mode and two-mode gates $\{\hat{R}_r^{(r)}, \ldots, \hat{R}_N^{(r)}\}$, demultiplexes the rth supermode $\hat{A}^{(r)} = \sum_m f_m^{(r)} \hat{a}_m$ to the rth (local) spatial bin. Operations up to $r = 2$ are shown, where local annihilation operators of the first and second bins are transformed to $\hat{A}^{(1)}$ and $\hat{A}^{(2)}$, respectively. Figure adapted from Yanagimoto et al. [1].

3.5 Wigner Negativity

Non-classicality measure: $\delta(\hat{\rho}) = \int d^{2N}\mathbf{r}\,(|W(\mathbf{r})| - W(\mathbf{r}))/2$.

4 Project Deliverables

4.1 Gaussian State MPS Construction

Implement Nüßeler et al. [2] algorithm:

1. Compute normal mode decomposition T s.t. $T^\dagger H T = D \oplus D$
2. Construct thermal MPO in normal modes: $\hat{\rho}_{\beta,\hat{\mathbf{b}}} = \bigotimes_k (1 - e^{-2\beta d_k}) \sum_{n_k} e^{-2\beta d_k n_k} |n_k\rangle\langle n_k|$
3. Decompose $T = \bar{U}\bar{S}\bar{V}^\dagger$ via Bloch-Messiah
4. Implement passive transformations using Reck/Clements decomposition: $\bar{U} = \prod_m (\prod_k \hat{B}_k(\theta_{m,k}) \hat{P}_k(\phi_{m,k}))$

5. Implement active transformation: $\hat{S}(\theta_k) = \exp(\theta_k(\hat{a}_k^2 - \hat{a}_k^{\dagger 2}))$
6. Apply $\hat{U} = \bar{U}\bar{S}\bar{V}^\dagger$ via MPO-MPO multiplication

4.2 Supermode Demultiplexing

Implement Yanagimoto et al. [1]:

1. For $r = 1, \ldots, s$, compute angles $\theta_m^{(r)}, \phi_m^{(r)}$ via iterative formula
2. Construct $\hat{R}^{(r)}$ from beam splitters and phase shifters
3. Apply $\hat{V}^{(s)}$ to MPS $|\Psi\rangle$ to obtain $|\Phi\rangle$
4. Extract reduced density matrix: $(\hat{\rho}_S)_{jj'} = \langle\Phi|(\bigotimes_{r=1}^{s} |j_r\rangle\langle j_r'| \otimes \bigotimes_{m=s+1}^{N} {}_m)|\Phi\rangle$
5. Compute Wigner function: $W(\alpha) = \frac{2}{\pi}\,\mathrm{Tr}[\hat{\rho}_S \hat{D}(\alpha)(-1)^{\hat{a}^\dagger\hat{a}}\hat{D}(-\alpha)]$

4.3 Time Evolution for Non-Gaussian States

Implement TEBD for Kerr nonlinearity:

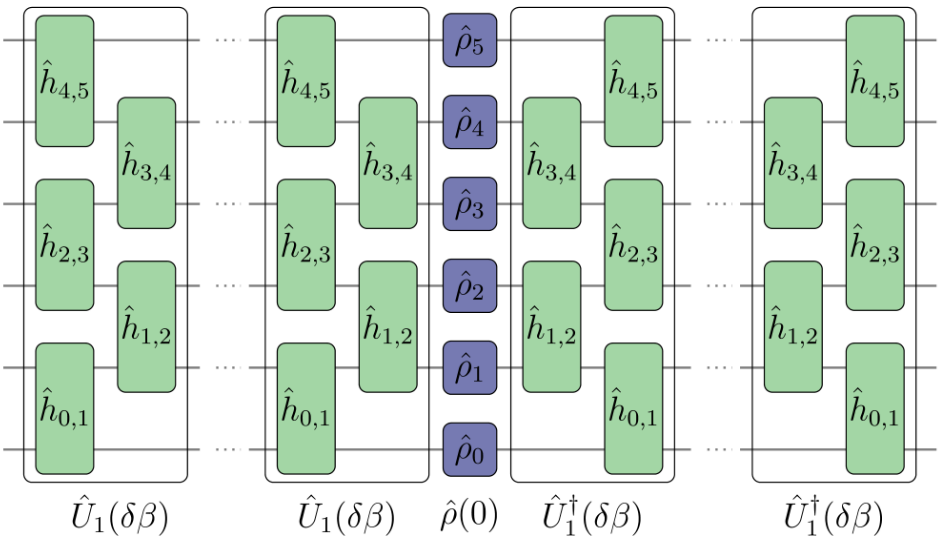

Figure 2: Tensor-network representation of density operator evolution via TEBD in imaginary time. The final state is obtained by successively applying the first-order propagator $\hat{U}_1(\delta\beta)$ to $\hat{\rho}$ from left and right. Each $\hat{U}_1(\delta\beta)$ consists of two layers comprising only two-site gates. For higher-order propagators, the number of layers increases. Figure adapted from Nüßeler et al. [2].

The Hamiltonian for a $\chi^{(3)}$ nonlinear waveguide is:

$$\hat{H} = -\frac{1}{2}\int dz(\hat{\phi}_z^\dagger \partial_z^2 \hat{\phi}_z + \hat{\phi}_z^{\dagger 2}\hat{\phi}_z^2)$$

TEBD implementation:

1. Trotterize: $\hat{U}(\delta t) \approx (\prod_{m \text{ odd}} \hat{D}_m)(\prod_{m \text{ even}} \hat{D}_m)(\prod_m \hat{S}_m)$
2. Apply gates sequentially, perform SVD truncation after each two-site gate
3. Compute observables: $\langle \hat{a}_m^\dagger \hat{a}_m \rangle$, $\mathrm{Tr}(\hat{\rho}_S^2)$, $\delta(\hat{\rho}_S)$

4.4 $\chi^{(2)}$ Hamiltonian Evolution

For $\chi^{(2)}$ nonlinear waveguides with fundamental harmonic (FH) and second harmonic (SH) modes:

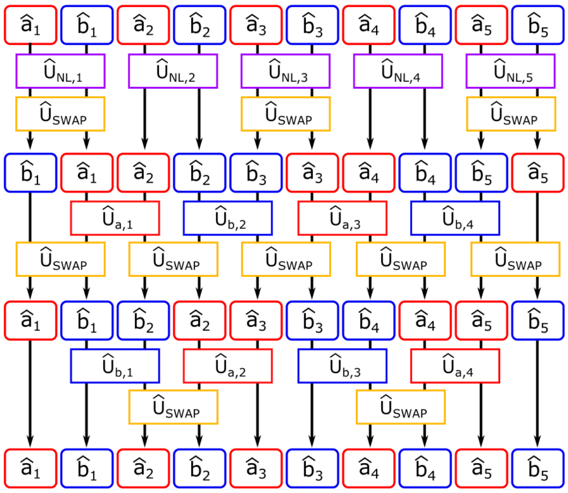

Figure 3: Discretized FH modes $\{\hat{a}_1, \ldots, \hat{a}_N\}$ and SH modes $\{\hat{b}_1, \ldots, \hat{b}_N\}$ are encoded on an MPS in alternating manner. Swap operations $\hat{U}_{\mathrm{SWAP}}$ bring distant modes together. Two-mode operations $\hat{U}_{a,m}$, $\hat{U}_{b,m}$, and $\hat{U}_{\mathrm{NL},m}$ implement FH dispersion, SH dispersion, and nonlinear three-wave mixing, respectively. Figure adapted from Yanagimoto et al. [1].

The $\chi^{(2)}$ Hamiltonian takes the form:

$$\hat{H} = \int dz \left(\hat{\phi}_z^\dagger \partial_z^2 \hat{\phi}_z + \hat{\psi}_z^\dagger \partial_z^2 \hat{\psi}_z + \hat{\phi}_z^{\dagger 2} \hat{\psi}_z + \mathrm{H.c.} \right)$$

4.5 Kerr Soliton Simulations

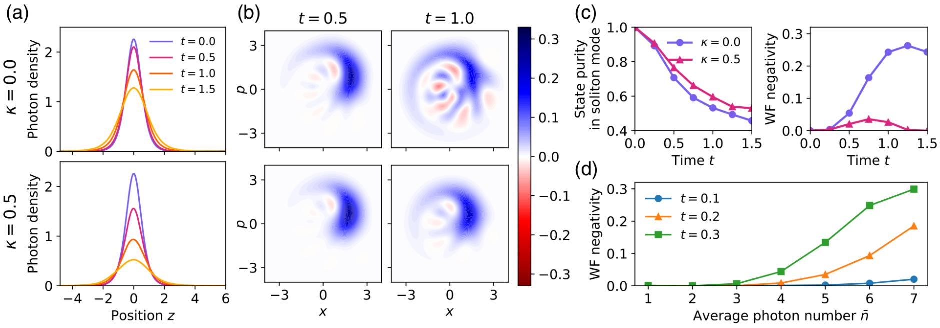

Figure 4: Quantum simulations of a pulse instantiated as a coherent-state sech soliton using MPS with bond dimension $\chi = 40$. Average photon number $\bar{n} = 3.0$ used with ($\kappa = 0.5$) and without ($\kappa = 0$) linear loss. MCWF with $M = 100$ quantum trajectories used for finite loss. (a) Time evolution of photon density distribution $\langle \hat{\phi}_z^\dagger \hat{\phi}_z \rangle$. (b) Wigner functions for reduced density matrix $\hat{\rho}_S$ for the sech supermode f^{sech}. Upper/lower rows: $\kappa = 0.0/0.5$. (c) Time evolution of purity $\mathrm{Tr}(\hat{\rho}_S^2)$ (left) and doubled volume of Wigner function negativity (right). (d) WF negativity at various times as function of $\bar{n}$ with $\chi = 50$. Figure adapted from Yanagimoto et al. [1].

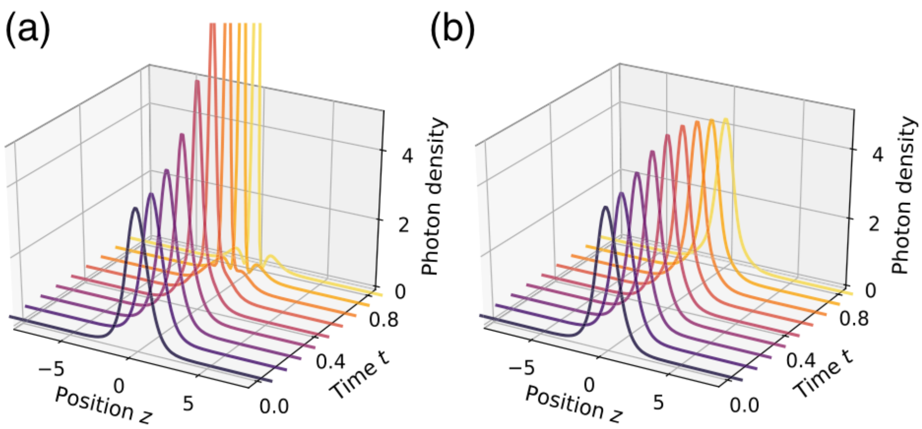

Figure 5: Time evolution of photon density distribution for a pulse instantiated as a coherent-state second-order soliton with mean photon number $\bar{n}^{(2\text{sech})} = 8$. (a) Classically expected dynamics. (b) Full quantum simulation under Hamiltonian Eq. (1) using MPS with bond dimension $\chi = 60$. Figure adapted from Yanagimoto et al. [1].

4.6 Benchmarking and Validation

- Compare with exact diagonalization for small systems ($N \leq 6$, $\chi \leq 20$)
- Reproduce Fig. 3 from Yanagimoto et al. (Wigner negativity for Kerr solitons)
  Reproduce Fig. 4 from Yanagimoto et al. (TDH comparison)
  Reproduce Fig. 5 from Yanagimoto et al. (second-order soliton dynamics)
  Reproduce Fig. 3 from Nüßeler et al. (floating-point operations scaling vs. temperature)

5 Timeline

Period	Activities
Community Bonding	Study MPS formalism, ITensors.jl API, CV quantum optics
Week 1-2	Literature review, environment setup, core tensor operations
Week 3-4	Implement Bloch-Messiah decomposition, Gaussian MPO construction
Week 5-6	Validate Gaussian MPO, implement benchmarks, reproduce Fig. 3 (Nüßeler)
Week 7-8	Implement supermode demultiplexing, Wigner function computation
Week 9-10	Implement TEBD for Kerr/$\chi^{(2)}$, reproduce Figs. 3-7 (Yanagimoto)
Week 11-12	Benchmarking, optimization, documentation, final report

Table 1: Proposed timeline for GSoC 2026

6 Complexity Analysis

Operation	Complexity	Notes
Gaussian MPO construction	$\mathcal{O}(N^3 + N^2\chi^2 d^2)$	N=modes, χ=bond dimension, d=cutoff
Supermode demultiplexing	$\mathcal{O}(sN^2\chi^2 d^2)$	s=number of supermodes
TEBD time evolution	$\mathcal{O}(N\chi^3 d^3)$	Per time step
$\chi^{(2)}$ evolution	$\mathcal{O}(N\chi^3 d^3)$	With swap operations

Table 2: Computational complexity of key operations

7 Lessons from Previous GSoC Contributors

7.1 QuantumClifford.jl Contributions

Previous successful GSoC contributors have established patterns I will follow:

- **Maria (#78)**: Implemented error-correcting code infrastructure. Key lessons:
  - Environment management is critical (using `] dev` vs `] add`)
  - Regular `] status` and `git status` checks prevent path confusion
  - Breaking work into small, focused PRs simplifies review
- **Trung (#89)**: Added encoding circuits and syndrome measurement. Key lessons:
  - Use `Rational` types (e.g., `1//2`) instead of floating point for exact arithmetic
  - Test logical operations by encoding random states and comparing circuit vs. algebraic methods
  - General-purpose functions preferred over code-specific implementations
- **Roy Ess (#289,#298,#356)**: Implemented qLDPC codes and BP-OSD decoder. Key lessons:
  - Large features delivered as multiple merges
  - Documentation of "Current state" vs "Left to do" sets clear expectations
  - Breaking changes require coordinated PRs
- **Shayan Pardis (QuantumCliffordGPU.jl)**: GPU acceleration. Key lessons:
  - Performance benchmarks across varying n essential for validation
  - GPU porting requires careful attention to memory layout and kernel design

7.2 Application to This Proposal

These lessons directly inform my implementation strategy:

1. Use `Rational` for all angle calculations to avoid floating-point accumulation errors
2. Implement comprehensive tests: encoding random states, comparing algebraic vs. circuit methods
3. Provide GPU acceleration path using CUDA.jl

4. Document current state and remaining work at each milestone
5. Maintain environment isolation with separate Project.toml
6. Reproduce key figures from Yanagimoto & Nüßeler papers as validation

8 Past Experience

- **Quantum Inspired Active Learning for Materials Discovery** (NQComp 2026)
  Wrote a paper on a quantum inspired active learning framework using amplitude encoding in Hilbert space and multi-observable uncertainty aggregation with non-commuting Hermitian operators for material science. Derived covariance aware acquisition functions modeling cross-property structure–property correlations. Across five independent trials and nine baselines (QBC, EI, BADGE, CoreSet, GP based methods), achieved 2.6–10.4% improvements in test R^2 and 25–35% reduction in sample complexity ($p < 0.01$). Complexity analysis shows scalable $O(NKd)$ performance under structured operators.
- **Lattice QCD and Critical Slowing Down**
  Supervisor: Pablo Morande (University of Cambridge)
  Implemented Hybrid Monte Carlo (HMC) and Metropolis–Hastings algorithms for lattice field simulations to analyze autocorrelation scaling and critical slowing through discretization of the harmonic oscillator.
- **Research Intern - JuliaSmoothOptimizers(JSO)**
  Supervisor: Dominique Orban (Polytechnique Montreal)
  Developing a structured pipeline to translate classical optimization problems into ADNLPModels and JuMP formulations for integration into OptimizationProblems.jl.
- **Collaborator - TagBot**
  Supervisor: Ian Butterworth
  Working on building Julia TagBot further with currently 17 merged prs and 2 open prs. Role of TagBot is to create tags, releases, and changelogs for your Julia packages when they're registered
- **Bayesian Modeling – sktime**
  Supervisor: Dr. Franz Kiraly (GCOS)
  Working on implementing Bayesian probabilistic estimators within the skpro ecosystem. Developing posterior predictive models, integrating uncertainty quantification and proper scoring rules (e.g. log-score, CRPS).
- **LFX Mentee – Linux Kernel**
  Contributed patches across filesystems(ext4, ntfs3, hpfs), memory management, and networking subsystems. Fixed use-after-free vulnerabilities and improved kernel build and error handling mechanisms.

8.1 Active Contributions to Gabs.jl

PR	Description	Status	Date
#94	No-Go Theorems section, Werner et al. (2001) reference	Merged	Jan 2026
#92	`embed` for `GaussianUnitary`, `GaussianChannel`, `GaussianLinearCombination`	Merged	Jan 2026
#91	Optional RNG arguments for deterministic testing	Merged	Jan 2026
#96	Heterodyne measurement interface	Open	Jan 2026
#104	Generalized `GaussianLinearCombination` to operators	Open	Mar 2026
#107	Quantum state tomography and performance docs	Open	Mar 2026

Table 3: Contributions to Gabs.jl

9 Commitment

I am completely free in the Summer and am prepared to give my full attention to the project.

In case of unavoidable circumstances leading to a temporary reduction in working hours, I will ensure to compensate for the time in the following weeks. I will communicate any such situations with my mentors well in advance to ensure smooth project progress.

For regular updates and discussions, I will be available via Teams, Zoom, and Google Meet in the Indian Standard Time Zone (GMT +5:30)

References

References

[1] R. Yanagimoto, E. Ng, L. G. Wright, T. Onodera, and H. Mabuchi, "Efficient simulation of ultrafast quantum nonlinear optics with matrix product states," *Optica*, vol. 8, no. 10, pp. 1306-1315, 2021.

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